Universität Zürich[UZH]

Center for Reproducible Science

Quantifying reproducibility - Results from a scoping review on reproducibility metrics and simulation studies into their real-world applicability

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Rachel Heyard*, Samuel Pawel, Joris Frese, Bernhard Voelkl, Hanno Würbel, Sarah McCann, Leonhard Held, Kimberley Wever, Helena Hartmann, Luise Townsin, and Stephanie Zellers

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What do we mean by “Reproducibility”?

Definition of reproducibility, for the purpose of this presentation:

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- *Computational reproducibility*:
  Using the *same data and code* to reproduce the original results.
- *Results reproducibility*:
  New data is collected and analysed using similar methods and materials.
- *Partial reproducibility*:
  Only parts of the study are reproduced (e.g., experiment, data collection, analysis), and results are compared.

The million-dollar question

— *When do results agree?*

— *What does it mean for study results to be consistent?*

Reproducibility Metrics Scoping Review | Question

Research questions:

- Which metrics have been used or suggested to quantify or assess which type of reproducibility?
- Which ones were suggested only, or have a clear implementation?
- How can they be interpreted and in which situations are they useful?
- What are assumptions and limitations?

Reproducibility Metrics Scoping Review | Question

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Reproducibility Metrics Scoping Review | Methods

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— Methodological papers – For reproducibility metrics **proposed, suggested or discussed**, a systematic search was conducted to identify relevant literature.

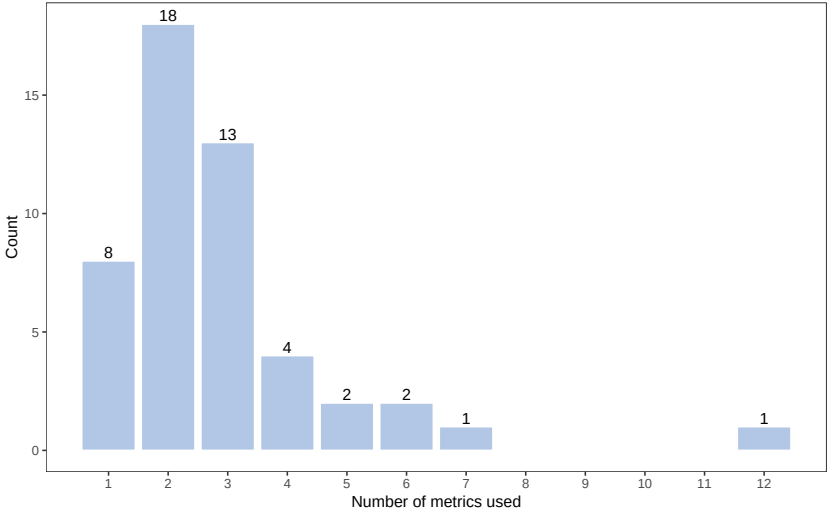
Application papers

- **49 records** of reproducibility projects across fields of research (based on OpenAlex):
  - Health and life sciences – 10 (20.4%)
  - Physical sciences – 6 (12.2%)
  - Social sciences – 33 (67.3%)

- Categorised into three project types:
  1. **Many phenomena, one study**: many original hypotheses tested, each hypothesis once;
  2. **One phenomenon, many studies**:one hypothesis is tested by many different teams;
  3. **Many phenomena, many studies**: many hypotheses are tested, each multiple times

Metrics used in application papers

- Various metrics were used to assess reproducibility:
  - Some compared **statistical significance**
  - Others compared **effect sizes**
  - Some used **meta-analyses**
  - Others relied on **subjective assessments**
  - *some used others*

Type of Metric	N (%)
Agreement in statistical significance	32 (65.3%)
Agreement in effect size	35 (71.4%)
Meta-analysis of study results	9 (18.4%)
Subjective assessment	14 (28.6%)

Number of metrics used in application papers